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# MUHAMMAD ASHIR

Efficient Machine Learning · Systems for AI · Multimodal Foundation Models · Embodied Perception

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## RESEARCH INTERESTS

I study how to make large multimodal models run where they are needed: on small, distributed, resource-constrained hardware. My work so far combines Mixture-of-Experts design, quantization, and distributed edge inference for vision-language models, measured on real single-board-computer clusters. For my Ph.D. I am interested in:

- **Efficient multimodal foundation models** — sparse Mixture-of-Experts, routing, parameter-efficient adaptation, knowledge distillation.
- **Systems for ML** — distributed and decentralized inference across heterogeneous edge devices; memory, communication, and scheduling bottlenecks.
- **Model compression** — low-bit quantization for vision-language models, especially for cross-attention and vision tokens.
- **Embodied AI** — efficient vision-language perception for robots operating in degraded or hazardous environments.

## EDUCATION

**M.S. in Software and Information Science (AI Course)** Apr 2025 – Mar 2027 (expected)

*Iwate Prefectural University, Graduate School of Software and Information Science — Iwate, Japan*

Thesis: *Optimization and Implementation of Vision-Language Models on Single-Board Computer Clusters*. Advisors: Prof. Hirokazu Madokoro, Prof. Stephanie Nix.

CGPA 3.71 / 4.00 (converted from official Japanese grades, credit-weighted; Excellent/Very Good → A, Good → B). Fully funded by the MEXT Japanese Government Scholarship.

Coursework: Machine Intelligence; Machine Learning and Applications; Brain Information Processing; Time Series Modeling and Prediction; Discrete System Simulation; Computational Mechanics; Sensing; HCI and Sensing.

**B.S. in Information Technology** Sep 2017 – Sep 2021

*Government College University Faisalabad — Faisalabad, Pakistan*

CGPA 3.20 / 4.00.

Coursework: Artificial Intelligence; Data Structures and Algorithms; Operating Systems; Computer Networks; Discrete Structures; Calculus and Analytical Geometry; Multivariable and Integral Calculus; Statistical Theory; Cloud Computing; Database Systems.

## PUBLICATIONS

- [J1] **M. Ashir**, H. Madokoro, and S. Nix. “Lightweight Edge AI for Vision-Language Models: A Mixture of Tiny Experts Approach for Distributed Inference on Single Board Computer Clusters.” *Manuscript in preparation for IEEE Access*, target submission Nov. 2026.
- [C2] **M. Ashir**, H. Madokoro, and S. Nix. “Mixture of Tiny Experts for Resource-Constrained Vision–Language Models on Single-Board Computer Clusters.” In *Proc. 12th International Conference on Computing and Artificial Intelligence (ICCAI 2026)*, ACM, Okinawa, Japan, Apr. 2026. **Best Presentation Award**; reviewers rated the work highly innovative.
- [C1] **M. Ashir**, H. Madokoro, and S. Nix. “Optimization and Implementation of Qwen-VL on Single Board Cluster Computers.” In *Proc. 26th SICE System Integration Division Annual Conference (SI2025)*, Hiroshima, Japan, pp. 203–204, Dec. 2025. **SI2025 Outstanding Presentation Award**.

## RESEARCH EXPERIENCE

**Graduate Researcher — Madokoro & Nix Laboratory** Sep 2024 – Present

*Iwate Prefectural University, Morioka, Japan*

**MoTE-VLM: a fully GPU-free vision-language model on a Raspberry Pi cluster [J1]**

- Designed and trained a 206M-parameter VLM (frozen MobileNetV3-Small encoder with 49 spatial tokens, 12 transformer blocks, hash-routed Mixture of Tiny Experts) whose entire inference runs CPU-only on a four-node Raspberry Pi 5 cluster, one expert per node.
- Two-stage training: caption alignment on COCO, then instruction tuning on answers distilled from a Qwen3-VL teacher. Reached **25.4 BLEU-4 / 70.4 CIDEr** on held-out COCO val2017 with no GPU at inference.
- Ran controlled studies: a parameter-free hash router outperformed a learned router (70.4 vs. 45.7 CIDEr; the learned router collapsed onto one or two experts), and an iso-compute dense baseline matched the mixture's quality (paired bootstrap), showing that the mixture's value is memory partitioning rather than accuracy.
- Characterized deployment trade-offs: INT8 post-training quantization made the model 2.7× smaller (309 MB) but only 1.1× faster on the Pi CPU (vs. 4.0× on x86); without a GPU, latency rose from 392 ms (1 node) to 630 ms (4 nodes), so distribution buys memory, not speed.

**Distributed tiny experts for Qwen2.5-VL-3B on single-board computers [C2]**

- Built a hybrid edge pipeline: Qwen2.5-VL-3B on a workstation GPU, with ~2.1M-parameter TinyExpert modules executing CPU-only on three Raspberry Pi 5 nodes over TCP/Gigabit Ethernet with top-1 softmax routing and a load-balancing loss.
- Reduced expert-stage latency from 87.24 ms (1 node) to 48.55 ms (3 nodes), a **1.80× speedup**; expert computation accounted for 81% of latency and expert load stayed within 15% of uniform.

**Quantized VLM for nuclear decommissioning imagery [C1]**

- Applied BitsAndBytes 8-bit quantization to Qwen2.5-VL-3B-Instruct, reducing memory allocation from 9,344 MB to 3,915 MB, and evaluated descriptions of low-light, blurry inspection images from the Fukushima Daiichi decommissioning site. Supported by JAEA Grant JPJA23O23813888.

Across all projects I carried out the literature review, system design, implementation, experiments, and LaTeX manuscript preparation; the move to Mixture-of-Experts came from my finding that dense deployment was infeasible on Pi hardware.

## HONOURS AND AWARDS

- **Best Presentation Award**, ICCAI 2026 (12th International Conference on Computing and Artificial Intelligence), Okinawa, Japan, Apr. 2026.
- **SI2025 Outstanding Presentation Award** (優秀講演賞), SICE System Integration Division, Feb. 2026.
- **MEXT Japanese Government Scholarship**, full tuition and monthly stipend for graduate study, Apr. 2025 – Mar. 2027.

## TECHNICAL SKILLS

**Programming:** Python, C++, Bash

**ML:** PyTorch, Hugging Face Transformers; vision-language models (Qwen2.5-VL, Qwen3-VL); Mixture-of-Experts (learned and hash routing, load balancing); quantization (BitsAndBytes, INT8 post-training); knowledge distillation

**Systems:** Raspberry Pi 5 clusters, TCP socket-based distributed inference, Linux administration, latency and memory profiling

**Tools:** Git/GitHub, LaTeX (IEEEtran, ACM sigconf)

## LANGUAGES

**English:** CEFR B2; research papers and conference talks in English. IELTS Academic scheduled for 14 Nov. 2026. **Japanese:** CEFR A2. **Urdu:** native.

## REFERENCES

**Prof. Hirokazu Madokoro** (primary advisor), Faculty of Software and Information Science, Iwate Prefectural University — hirokazu_m@iwate-pu.ac.jp

**Prof. Stephanie Nix** (co-advisor), Faculty of Software and Information Science, Iwate Prefectural University — nix_s@iwate-pu.ac.jp